

Tool Condition Monitoring in CNC Milling using Sensor Data and ML

Manufacturing & Data Science Project | B.Tech Mechanical Engineering, IIT Kharagpur | One-Page Project Report

Objective

CNC controllers log motor-level signals (current, voltage, velocity) during every job. This project asks whether a worn vs. unworn cutting tool can be detected purely from these signals, without stopping the machine for manual inspection — the basis of real Tool Condition Monitoring (TCM) systems used in Industry 4.0 to cut downtime and scrap.

Dataset

CNC Mill Tool Wear Dataset (University of Michigan, SMART Lab). 18 milling experiments engraving an “S” pocket into a wax block, each varying feed rate (3–20 mm/s) and clamp pressure (2.5–4.0 bar), using either a fresh or visibly worn tool (8 unworn / 10 worn). Each experiment logs 48 channels (position, velocity, current, voltage, power per axis) at 100 ms resolution. Source mirror: github.com/SaeedShurrab/Tool-Wear-Detection-in-CNC-Milling-Operations.

Methodology — Two Complementary Approaches

	Approach A — Per-Experiment Signature	Approach B — Row-Level Classification
Features	Time-domain stats (mean, RMS, std, peak-to-peak, skew, kurtosis) + FFT features per signal, per experiment	Raw 17 sensor channels used directly, one row per 100 ms sample during active cutting
Samples	18 (one per experiment)	~17,500 rows from 18 experiments
Validation	Leave-One-Out CV, with feature selection nested inside each fold to avoid leakage	Stratified, group-aware split — entire experiments held out, never split across train/test
Models	Logistic Regression, Random Forest	Random Forest, XGBoost

Results

Metric	Approach A (LOOCV)	Approach B (held-out experiments)
Accuracy	0.50 (chance level)	0.40–0.42 ($\leq$ majority-class baseline of 0.67)
ROC-AUC	—	0.52–0.53 (chance $\approx$ 0.50)
Top-ranked signals	X/Y-axis current & voltage means	X/Y-axis & spindle output current

Headline finding, stated directly: neither approach reliably beats a trivial baseline on genuinely unseen experiments. Both consistently point to axis/spindle current and voltage as the most separable channels — physically sensible, since cutting load shows up first in motor current — but that ranking does not translate into a model that generalizes across experiments.

Why the Signal Is Weak, and What Would Fix It

- **Soft workpiece material.** Wax shows far smaller wear-induced load changes than metal would.
- **Operating-point confounding.** Feed rate and clamp pressure change signal magnitude independently of, and alongside, tool condition across experiments.
- **Coarse binary label.** `tool_condition` has no continuous flank-wear (VB) measurement.
- **Only 18 independent trials** exist — row-level features cannot manufacture new independent data.
- **Fix:** repeat on a metal dataset with continuous VB (e.g. NASA/UC Berkeley milling data) as a regression task; control for feed rate/clamp pressure explicitly; add a true force/vibration/AE sensor.

Takeaway

The deliverable here is methodological rigor, not an inflated accuracy figure: nested feature selection, Leave-One-Out CV for a small sample, group-aware train/test splitting to prevent leakage, and an explicit baseline comparison that exposed a result worth reporting honestly rather than hiding behind a favorable split.